

20 Kaggle Datasets for Regression Machine Learning

Regression datasets are used to predict **continuous numerical values** such as prices, sales, costs, temperatures, scores, or demand forecasts. Many of these are popular benchmarks in machine learning courses and competitions. [\[kaggle.com\]](#), [\[kaggle.com\]](#)

#	Dataset	URL	Short Description
1	House Prices: Advanced Regression Techniques	https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques	Predict residential house sale prices from property features.
2	California Housing Prices	https://www.kaggle.com/datasets/camnugent/california-housing-prices	Predict median house values in California districts. [kaggle.com] , [jmyao17.github.io]
3	Medical Cost Personal Dataset	https://www.kaggle.com/datasets/mirichoi0218/insurance	Predict health insurance charges based on demographics and lifestyle factors.
4	NYC Airbnb Open Data	https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data	Predict Airbnb listing prices from location and property attributes.
5	BigMart Sales Prediction	https://www.kaggle.com/datasets/brijbhushannanda1979/bigmart-sales-data	Forecast product sales across retail stores.
6	Walmart Store Sales Forecasting	https://www.kaggle.com/competitions/walmart-recruiting-store-sales-forecasting	Predict future store sales using historical data.
7	Bike Sharing Demand	https://www.kaggle.com/competitions/bike-sharing-demand	Predict hourly bicycle rental demand.
8	Rossmann Store Sales	https://www.kaggle.com/competitions/rossmann-store-sales	Forecast daily sales for Rossmann stores.
9	Restaurant Revenue Prediction	https://www.kaggle.com/competitions/restaurant-revenue-prediction	Predict annual restaurant revenue.
10	Fish Market Dataset	https://www.kaggle.com/datasets/aungpyaeap/fish-market	Predict fish weight from physical measurements. [github.com]

11	Life Expectancy (WHO)	https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who	Predict life expectancy using health and socioeconomic indicators. [github.com]
12	Tomato Prices Dataset	https://www.kaggle.com/datasets/srinuti/residential-power-usage-3years-data-timeseries	Predict agricultural commodity prices.
13	Used Car Dataset	https://www.kaggle.com/datasets/nehalbirla/vehicle-dataset-from-cardekho	Predict used-car market prices.
14	Diamond Prices	https://www.kaggle.com/datasets/shivam2503/diamonds	Predict diamond prices using cut, color, and clarity attributes.
15	Student Performance	https://www.kaggle.com/datasets/spscientist/students-performance-in-exams	Predict exam scores from demographic and educational factors.
16	Concrete Compressive Strength	https://www.kaggle.com/datasets/maajdl/yeh-concretedata	Predict concrete strength from material composition.
17	Energy Efficiency Dataset	https://www.kaggle.com/datasets/elikplim/car-evaluation-data-set	Predict building heating and cooling loads.
18	Automobile Price Prediction	https://www.kaggle.com/datasets/toramky/automobile-dataset	Predict vehicle prices using technical specifications.
19	Weather History Dataset	https://www.kaggle.com/datasets/muthuj7/weather-dataset	Predict temperature and weather-related numerical values.
20	Real Estate Valuation Dataset	https://www.kaggle.com/datasets/quantbruce/real-estate-price-prediction	Predict real-estate prices from property attributes.

20 Hugging Face Datasets for Regression Machine Learning

Hugging Face contains many regression datasets where the target is a **continuous score** rather than a class label. These datasets are commonly used for NLP, semantic similarity, scoring, forecasting, and numerical prediction tasks. [\[huggingface.co\]](https://huggingface.co), [\[huggingface.co\]](https://huggingface.co)

#	Dataset	URL	Short Description
1	STS-B (Semantic Textual Similarity Benchmark)	https://huggingface.co/datasets/SetFit/stsb	Predict similarity scores (0–5) between sentence pairs. [huggingface.co]
2	GLUE STS-B	https://huggingface.co/datasets/glue/viewer/stsb	Benchmark dataset for sentence similarity regression.
3	Essay Scoring Dataset	https://huggingface.co/datasets/Ericwang/ASAP	Predict essay quality scores assigned by human graders.
4	CommonLit Readability Prize	https://huggingface.co/datasets/commonlit/commonlitreadabilityprize	Predict text readability scores.
5	Airbnb Price Prediction	https://huggingface.co/datasets/huggingface-course/airbnb-regression	Predict Airbnb rental prices.
6	California Housing	https://huggingface.co/datasets/scikit-learn/california_housing	Predict California housing prices. [keras.io] , [inria.github.io]
7	Ames Housing	https://huggingface.co/datasets/jlh/ames_housing	Predict residential property prices.
8	Wine Quality	https://huggingface.co/datasets/codesignal/wine-quality	Predict numerical wine quality ratings.
9	Medical Insurance Cost	https://huggingface.co/datasets/TrainingDataPro/insurance_cost	Predict insurance charges.
10	Student Performance Regression	https://huggingface.co/datasets/mstz/student_performance	Predict student exam scores.
11	Boston Housing (Archived)	https://huggingface.co/datasets/scikit-learn/boston_housing	Predict housing prices using neighborhood features.
12	Energy Efficiency	https://huggingface.co/datasets/uoft-cs/energy_efficiency	Predict heating and cooling loads.
13	Concrete Strength	https://huggingface.co/datasets/Matthijs/concrete_compressive_strength	Predict concrete compressive strength.
14	Abalone Age Prediction	https://huggingface.co/datasets/scikit-learn/abalone	Predict abalone age from physical measurements.
15	Fish Weight Regression	https://huggingface.co/datasets/yanirs/fish-market	Predict fish weight from dimensions.

16	Used Car Prices	https://huggingface.co/datasets/ali7d/used_cars	Predict used-car prices.
17	House Rent Prediction	https://huggingface.co/datasets/house-rent-prediction	Predict monthly rental prices.
18	Employee Salary Prediction	https://huggingface.co/datasets/salary-prediction	Predict employee salaries from profile attributes.
19	Product Demand Forecasting	https://huggingface.co/datasets/product-demand-forecasting	Forecast continuous product demand.
20	Real Estate Valuation	https://huggingface.co/datasets/real-estate-valuation	Predict real-estate valuation prices.

Best Kaggle Regression Datasets for Beginners

1. House Prices
2. California Housing Prices
3. Medical Cost Personal Dataset
4. Fish Market
5. Diamond Prices

Best Hugging Face Regression Datasets for NLP

1. STS-B
2. GLUE STS-B
3. CommonLit Readability Prize
4. ASAP Essay Scoring
5. Semantic Similarity Benchmarks

Best Regression Datasets for Deep Learning

1. House Prices
2. California Housing
3. NYC Airbnb

4. Energy Efficiency

5. Ames Housing

These 40 datasets cover:

- House Price Prediction
- Sales Forecasting
- Demand Forecasting
- Cost Estimation
- Readability Scoring
- Semantic Similarity Regression
- Salary Prediction
- Insurance Cost Prediction
- Real Estate Valuation
- Engineering and Energy Regression Tasks.